An Intelligent AI-Driven Decision Support System for Personalized Income Tax Compliance and Advisory in Sri Lanka

R26-DS-004



Project Proposal Report

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March 2026

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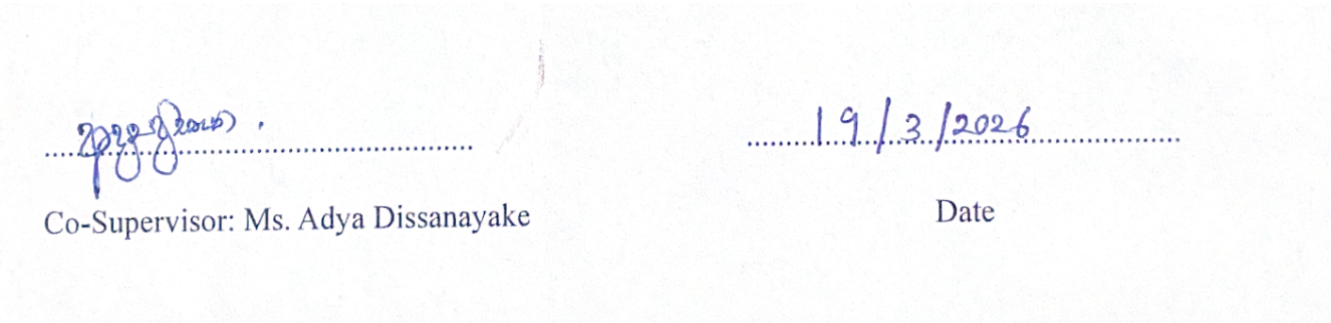
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Sri Lanka's personal income tax system requires individuals to declare their total income and submit annual tax returns, yet many taxpayers face difficulties in identifying tax-relevant income accurately from raw financial records. Personal transaction data are often heterogeneous, noisy, and semantically ambiguous, making manual interpretation time-consuming and error-prone. Existing tax portals, accounting tools, and prior machine learning studies provide partial support for filing or transaction categorization, but they do not offer an explainable, confidence-aware framework that semantically interprets personal financial transactions and infers taxable income in the Sri Lankan context.

This research proposes an AI-driven Financial Transaction Semantic Reasoning and Explainable Taxable Income Inference framework to support personalized income tax compliance. The proposed system transforms raw transaction records into a normalized machine-readable representation, applies semantic transaction understanding models to identify meaningful financial events, maps these events to tax-relevant categories, and generates explainable taxability decisions with supporting evidence. To improve practical reliability, the framework also incorporates uncertainty estimation, novelty detection, and human-in-the-loop review for ambiguous or previously unseen transaction patterns.

The methodology combines transaction preprocessing, semantic modeling, economic event inference, tax-rule-grounded reasoning, and empirical validation using classification, estimation, calibration, and generalization metrics. Expected outcomes include a structured transaction dataset, a semantic classification model, an explainable taxability reasoning module, a confidence-aware taxable income inference mechanism, and an experimentally validated prototype that can improve interpretability and reduce manual review burden. The proposed research contributes both academic value and practical significance by addressing an important gap in intelligent tax decision-support systems for Sri Lanka.

Keywords: Financial transaction semantics, taxable income inference, explainable AI, tax compliance, Sri Lanka

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

API – Application Programming Interface

BERT – Bidirectional Encoder Representations from Transformers

Bi-LSTM – Bidirectional Long Short-Term Memory

ECE – Expected Calibration Error

FinBERT – Financial Bidirectional Encoder Representations from Transformers

F1-Score – Harmonic Mean of Precision and Recall

IRD – Inland Revenue Department

LLM – Large Language Model

MC-Dropout – Monte Carlo Dropout

ML – Machine Learning

NFR – Non-Functional Requirement

OOD – Out-of-Distribution

PDF – Portable Document Format

RAG – Retrieval-Augmented Generation

RMSE – Root Mean Square Error

SDG – Sustainable Development Goal

UI – User Interface

USP – Unique Selling Proposition

WBS – Work Breakdown Structure

1. INTRODUCTION

1.1 Background and Context

1.1.1 Sri Lankan personal income tax compliance context

Income taxation plays a central role in public revenue mobilization and fiscal stabilization in Sri Lanka. Recent macro-fiscal reports indicate that tax revenue performance has improved as part of broader revenue-based fiscal reforms, with tax revenue increasing significantly in recent years [1], [2]. At the same time, the effectiveness of tax administration depends not only on policy reform, but also on the ability of taxpayers to understand, report, and comply with their obligations accurately and on time. In Sri Lanka, income tax is administered under the Inland Revenue Act, No. 24 of 2017 and its amendments, while the Inland Revenue Department (IRD) provides online facilities for taxpayer registration, filing returns, checking balances, and making payments [3], [4].

For individual taxpayers, return filing remains a legal obligation even in cases where tax is deducted at source through employment-related mechanisms. Current IRD guidance for individual income tax returns states that persons are required to submit their tax returns electronically, and the return process requires the declaration of total income and other relevant particulars for the year of assessment [5], [6]. This creates an important compliance reality in which employer-side deductions and individual-side filing obligations coexist, making personal awareness and correct reporting essential to lawful tax compliance.

1.1.2 Challenges faced by salaried taxpayers and tax administrators

Despite the availability of employer-based tax deductions and expanding digital tax services, many salaried individuals may still misunderstand the extent of their own filing responsibilities. In practical settings, taxpayers often interpret advance deductions as full completion of their obligations, especially when they do not have strong tax literacy, do not actively engage with the IRD portal, or receive income from multiple sources beyond a primary salary. This creates a compliance environment in which filing behavior may become passive, delayed, or incomplete.

The problem is not limited to taxpayers alone. Tax compliance and review processes also affect tax administrators, accountants, and auditors, who may need to assess whether reported or unreported financial inflows represent salary, freelance income, gifts, reimbursements,

loans, capital gains, or other economically distinct events. Your project concept correctly identifies that manual review of transaction records is time-consuming, error-prone, and highly dependent on statement format and language variation, especially in Sri Lankan banking contexts where English, Sinhala, and Tamil may appear in mixed or abbreviated forms. As a result, both taxpayers and tax professionals face practical barriers in producing transparent, explainable, and timely income-tax compliance decisions.

1.1.3 Limitations of current filing, advisory, and transaction-review practices

Sri Lanka has made visible progress in digital tax administration through e-services that support registration, online filing, payments, refunds, objections, and transaction history access [4]. These services improve accessibility and administrative reach, but they do not by themselves solve the cognitive and analytical burden placed on individuals when they must determine what income to report, how to interpret mixed financial records, and whether a transaction is taxable, exempt, non-taxable, or requires further review. In other words, current platforms are valuable for **processing** compliance, but they do not sufficiently support **reasoning** about compliance.

This limitation becomes more significant when taxpayers have multiple income channels, inconsistent transaction descriptions, informal transfers, or records distributed across different banks and financial platforms. Existing tax portals and form-based filing systems are primarily designed to receive declarations rather than semantically interpret raw personal financial evidence. Likewise, conventional manual review by accountants or auditors does not scale well, and outcomes may differ depending on the reviewer's experience, interpretation, and time constraints. These issues suggest that the core challenge is not merely digitizing filing, but enabling reliable, explainable, and context-aware interpretation of financial activity for tax purposes.

1.1.4 Rationale for an intelligent AI-driven tax decision support system

Advances in artificial intelligence, machine learning, and natural language processing create an opportunity to move beyond static filing interfaces toward intelligent tax decision support. A system that can ingest transaction descriptions, amounts, dates, and surrounding context may help infer the underlying economic meaning of a financial event, estimate its probable tax relevance, and generate transparent reasoning to support human understanding. This is especially relevant in Sri Lanka, where multilingual transaction text, heterogeneous bank

statement formats, and varying taxpayer awareness create a high-friction compliance environment.

Rather than replacing tax professionals or legal interpretation, such a system can function as an assistive layer that improves early compliance, strengthens taxpayer understanding, and reduces avoidable ambiguity before formal return submission. It can also help identify uncertain or suspicious cases that require human review instead of forcing binary decisions on incomplete evidence. In this sense, an AI-driven tax platform is not simply an automation tool; it is a decision-support mechanism that can promote proactive compliance, better documentation, and more transparent reporting behavior.

1.1.5 Overall project context and component alignment

This research proposes an intelligent AI-driven decision support system for personalized income tax compliance and advisory in Sri Lanka. The overall system is designed as a multi-component platform covering financial transaction understanding, tax strategy optimization, tax advisory conversation support, and personalized recommendation. Your project notes define these four components as: financial transaction semantics and taxable income inference, learning-based tax strategy optimization and explainable decision support, a knowledge-grounded tax advisory chatbot, and a personalized recommendation and predictive impact engine.

Within that larger architecture, the financial transaction reasoning component is foundational because it transforms raw financial evidence into structured tax-relevant knowledge that can be used by the other components. In its revised form, this component is better understood not as a simple taxable/non-taxable classifier, but as a semantic reasoning framework that performs transaction understanding, economic event identification, tax-rule mapping, explainable taxability reasoning, uncertainty detection, and evidence generation. That framing is more consistent with the overall system vision of supporting personalized, explainable, and proactive tax compliance in a real-world Sri Lankan context.

1.1.6 SDG and research-cluster alignment

This research aligns naturally with broader goals of strengthening institutions, improving access to accountable public services, and using digital innovation to support inclusive governance. In particular, it is relevant to **Sustainable Development Goal 16**, especially in relation to effective, accountable, and transparent institutions, and to **Sustainable**

Development Goal 9, through the use of intelligent digital infrastructure and innovation in public-facing service delivery. Conceptually, it also fits computing research themes related to artificial intelligence, intelligent information systems, financial technology, decision support, and applied data science. Because the proposed system addresses a legally grounded, high-impact public-administration problem using explainable AI and data-driven reasoning, its alignment appears substantive rather than forced.

1.2 Literature Review

1.2.1 Digital tax services and existing taxable income products

Existing digital tax and accounting platforms demonstrate that tax and financial workflows are increasingly supported by software, but most current systems emphasize filing support, rule-based categorization, and user guidance rather than deep semantic reasoning over raw personal financial evidence. In Sri Lanka, the Inland Revenue Department provides e-services for taxpayer registration, filing returns, checking balances and transaction history, making payments, requesting refunds, and lodging appeals or objections, showing that the national tax environment is becoming more digitalized [4]. However, these services are mainly designed to facilitate tax administration processes rather than infer the semantic meaning or taxability of raw banking transactions automatically. Similarly, commercial systems such as QuickBooks and Zoho Books support automated or rule-based transaction categorization, while TurboTax offers AI-assisted tax guidance and question answering [7]–[9]. These tools are useful for bookkeeping or assisted filing, but they do not provide auditor-style evidence chains that connect a raw transaction to an economic event, a tax rule, and a calibrated taxability explanation. This reveals an important distinction between workflow automation and explainable tax reasoning, which is central to the proposed research.

1.2.2 Financial transaction classification and semantic understanding

Academic work on financial transaction classification shows that transaction text can be modeled successfully using modern machine learning and transformer-based methods, but the literature has mostly focused on categorization rather than tax reasoning. Toran et al. proposed a scalable weakly supervised bank transaction classification pipeline that combines preprocessing, embeddings, heuristic label generation, and neural classification to reduce dependence on expensive manual labels [10]. Busson et al. introduced a transformer-based hierarchical classification model for financial transactions, showing that contextual embeddings and taxonomy-aware attention can improve transaction categorization [11]. Ta et

al. also highlighted that bank transaction classification remains underexplored because access to financial data is restricted, and they showed that short transaction descriptions require careful language-specific preprocessing due to brevity, noise, and domain-specific wording [12]. Together, these studies show that transaction text classification is feasible and increasingly sophisticated, but they largely stop at assigning a category label rather than identifying the underlying economic event and producing a tax-grounded explanation of why a transaction should be treated in a certain way.

1.2.3 Weak supervision for scarce financial labels

A major challenge in transaction understanding is the lack of large, high-quality labeled datasets, especially in privacy-sensitive financial settings. Snorkel and related weak-supervision approaches were developed precisely to address this bottleneck by allowing labeling functions, heuristics, and domain rules to generate probabilistic training labels instead of relying entirely on manual annotation [13]. In the transaction domain, Toran et al. explicitly applied this idea by using heuristics and domain knowledge to create scalable bank transaction labels for model training [10]. This is highly relevant to the proposed Sri Lankan tax setting, where manually annotated multilingual transaction corpora are unlikely to be available at the scale needed for supervised learning. Weak supervision therefore provides a practical foundation for bootstrapping a tax-oriented transaction reasoning system, especially when paired with synthetic data generation and expert review. However, weak labels alone do not solve the problem of explainability, nor do they guarantee that a model's final prediction is legally meaningful or sufficiently transparent for tax advisory use.

1.2.4 Sequence and temporal modeling of transaction behavior

Several recent studies argue that financial transactions should not be treated only as isolated text records, because important meaning is often distributed across time, repetition patterns, and sequence context. Garuti et al. note that transactional data are naturally organized as time-dependent sequences containing heterogeneous fields such as dates, amounts, and categories, and they argue that many conventional approaches do not fully exploit temporal patterns and the information richness of individual transactions [14]. Their UniTTab transformer architecture was proposed specifically to model heterogeneous transaction sequences end-to-end [14]. Likewise, Noels et al. explored semantic embeddings for financial ledger descriptions, demonstrating that contextual representations can capture meaningful relationships between transaction descriptions and financial events. These findings support

the proposed pipeline, because taxability often depends not only on one transaction description, but also on repeated inflows, periodic salary-like behavior, linked transfers, or broader account history patterns.

1.2.5 Explainability, confidence calibration, and trustworthy AI in finance

Explainability is especially important in finance because high-stakes decisions require transparency, justification, and user trust. Lundberg and Lee introduced SHAP, a widely used framework for interpreting machine learning predictions by measuring the contribution of each feature to the final model output [16]. In parallel, Guo et al. demonstrated that modern neural networks are often poorly calibrated and may produce overconfident predictions, highlighting the need for explicit calibration techniques such as temperature scaling [17]. In financial applications, explainable AI helps ensure that automated decisions remain transparent and interpretable to both domain experts and end users, and recent reviews emphasize that finance is a critical domain where decision-making transparency is especially important [18]. However, most explainability methods focus on feature importance rather than generating structured reasoning chains that connect transaction semantics, economic events, and applicable tax rules.

1.2.6 Human-in-the-loop learning, active learning, and novel transaction discovery

Because financial transaction ecosystems evolve continuously, practical systems must be able to adapt to new patterns that may not exist in the original training data. Active learning has been widely studied as a method for reducing annotation effort while maintaining model performance, particularly in text classification settings [19]. Meanwhile, research on novel class discovery focuses on identifying previously unseen categories in unlabeled data while learning from existing labeled classes [20]. These approaches are relevant to financial transaction analysis because new transaction types, abbreviations, and financial behaviors frequently appear in real-world data. A human-in-the-loop learning framework that routes uncertain transactions to expert review can therefore improve both system adaptability and trustworthiness.

1.2.7 Summary of limitations in existing literature

Overall, the literature provides strong foundations for transaction classification, weak supervision, temporal representation learning, explainable AI, and uncertainty-aware learning. However, several limitations remain when these strands are viewed from the perspective of personalized income tax compliance in Sri Lanka. First, most prior work

focuses on classification or prediction rather than taxability reasoning. Second, many studies rely on proprietary or non-public datasets from specific banking environments, which limits evidence of generalization across banks, countries, and languages. Third, even when explainability is discussed, the explanation is often a feature-importance perspective rather than a structured evidence chain that links transaction semantics to economic events and tax rules. Fourth, the reviewed systems are not tailored to multilingual bank-statement text environments where Sinhala, Tamil, and English may appear in mixed form. Finally, existing financial software products support filing, bookkeeping, and categorization but do not provide decision-support workflows that produce taxable amount estimates together with confidence and explainable reasoning. These limitations motivate the proposed research direction of **Financial Transaction Semantic Reasoning and Explainable Taxable Income Inference**.

1.3 Research Gap

1.3.1 Gap in existing financial transaction classification research

Existing studies in financial transaction analysis have shown that machine learning models can classify transaction descriptions into predefined categories with strong performance using weak supervision, transformer-based embeddings, and hierarchical classification architectures [10], [11]. These studies demonstrate that transaction text can be modeled effectively despite noise and limited labeled data. However, their primary focus remains on assigning category labels such as merchant type or transaction class rather than identifying the underlying economic event behind a transaction. As a result, current research is strong in **transaction categorization**, but still limited in **economic interpretation** and **tax-oriented semantic reasoning**, which are essential for personalized income tax decision support.

1.3.2 Gap in explainability and tax decision support

Research in explainable AI has emphasized that transparency is especially important in high-stakes domains such as finance, where decisions must be understandable, justifiable, and trustworthy [18]. Likewise, studies on confidence calibration show that modern neural networks can be overconfident, meaning that prediction scores alone are not sufficient for trustworthy decision-making [17]. However, existing explainability methods usually focus on feature importance or local model interpretation rather than generating a structured reasoning chain that connects a raw transaction to an inferred economic event, a candidate tax rule, and a justified taxable amount. In other words, the literature addresses **why a model predicted**

something more often than **how a tax-relevant conclusion should be reasoned step by step**. This creates a clear gap between explainable machine learning and truly explainable tax decision support.

1.3.3 Gap in existing digital tax and financial support systems

Existing digital tax and financial software platforms support activities such as filing, bookkeeping, registration, transaction rule setting, and AI-assisted guidance, but they do not provide a full reasoning layer for personalized taxability inference. In Sri Lanka, the Inland Revenue Department's e-services allow taxpayers to register, file returns, check balances and transaction history, make payments, request refunds, and lodge appeals or objections [4]. These are important administrative capabilities, but they are mainly designed to facilitate tax processes rather than semantically interpret personal financial records and explain why a transaction should be considered taxable, non-taxable, partially taxable, or uncertain. Similarly, mainstream commercial platforms focus on workflow automation and financial categorization rather than evidence-based tax reasoning. Therefore, there remains a gap between **digital tax service availability** and **intelligent taxability reasoning support**.

1.3.4 Gap in the Sri Lankan personalized income tax compliance context

The gap becomes more significant in the Sri Lankan context, where transaction understanding for tax purposes is complicated by heterogeneous bank statement formats, multilingual transaction descriptions, and the need to interpret personal financial inflows under local compliance requirements. Prior research on financial transaction classification and explainable AI is generally not designed for Sri Lankan tax administration conditions, nor for multilingual financial text involving Sinhala, Tamil, and English mixtures. Furthermore, existing studies do not sufficiently address the practical need to transform noisy, cross-bank personal transaction records into **economic event identification, tax-rule mapping, taxability reasoning, taxable amount estimation, and calibrated confidence with supporting evidence**. This indicates a clear contextual and methodological gap for an intelligent, explainable, and locally relevant decision-support framework for personalized income tax compliance in Sri Lanka.

1.3.5 Identified research gap of the proposed study

Accordingly, the main research gap addressed by this study is the lack of a **robust, explainable, and context-aware framework** that can move beyond financial transaction classification toward **Financial Transaction Semantic Reasoning and Explainable Taxable**

Income Inference. Existing work provides useful foundations in classification, weak supervision, and explainability, but none sufficiently integrates semantic transaction understanding, economic event inference, tax-rule mapping, uncertainty-aware reasoning, and evidence generation within a Sri Lankan personalized tax-compliance setting. This gap justifies the proposed research as a novel bridge between machine learning-based financial transaction understanding and explainable tax decision support.

1.4 Research Problem

Although digital tax services and financial software systems are available, they do not adequately help individual taxpayers understand the tax relevance of their personal financial transactions. Existing approaches mainly support filing, bookkeeping, or basic transaction categorization, but they do not provide explainable reasoning that identifies the underlying economic event of a transaction, maps it to relevant tax rules, and justifies whether the amount is taxable, non-taxable, partially taxable, or uncertain [4], [11], [18].

This problem is more challenging in Sri Lanka due to heterogeneous bank statement formats, multilingual transaction descriptions, limited labeled financial data, and the lack of intelligent tax-support systems tailored to local compliance requirements. As a result, taxpayers may struggle to accurately understand, report, and justify their taxable income, while manual review remains time-consuming, error-prone, and difficult to scale [10].

Therefore, the research problem addressed in this study is the lack of an explainable, context-aware, and intelligent framework that can semantically analyze personal financial transactions and infer taxable income with supporting evidence for personalized income tax compliance in Sri Lanka.

1.4.1 System comparison (Existing Systems vs Proposed System)

Table 1 - System comparison

Feature	**Tax Portals** IRD e-Services / RAMIS	**Accounting Software** QuickBooks Online, Zoho Books	**Tax-assistance software** TurboTax / Intuit Assist	**Existing ML research** Toran et al. (2023), Busson et al. (2023)	**Proposed System** Our Solution
Filing support	**Yes**	**Partial**	**Yes**	**No**	**Yes**
Transaction classification	**No**	**Basic**	**Partial**	**Yes**	**Yes**
Explainable tax reasoning	**No**	**No**	**No**	**No**	**Yes**
Taxable amount inference	**No**	**Limited**	**No**	**Limited**	**Yes**
Confidence score	**No**	**No**	**No**	**Partial**	**Yes**
Sri Lankan context	**Partial**	**No**	**No**	**No**	**Yes**

2. OBJECTIVES

The objectives of this component define the intended research and development path for the proposed **Financial Transaction Semantic Reasoning and Explainable Taxable Income Inference** framework. They are designed to ensure that the study is not limited to transaction categorization alone, but instead addresses the full reasoning pipeline required for tax-relevant financial interpretation in the Sri Lankan context. In line with the identified research problem, the objectives focus on semantic transaction understanding, economic event inference, tax-rule-grounded reasoning, uncertainty-aware taxable income estimation, and empirical validation through measurable performance indicators.

2.1 Main Objective

To design, develop, and validate an explainable AI-driven framework that semantically analyzes personal financial transactions, identifies tax-relevant economic events, and infers taxable income with supporting evidence and calibrated confidence in order to support personalized income tax compliance in Sri Lanka.

2.2 Specific Objectives

SO1: Develop a transaction data representation and normalization pipeline

To design and implement a preprocessing and normalization pipeline capable of transforming heterogeneous personal financial transaction records into a unified machine-readable representation by extracting and standardizing key attributes such as transaction descriptions, amounts, dates, directions of flow, and contextual metadata.

This objective addresses the practical challenge that financial records may originate from different banks, statement formats, and inconsistent description structures. The pipeline should support noisy and abbreviated transaction text and prepare the data for downstream semantic analysis. Where possible, it should also account for multilingual or mixed-language transaction descriptions relevant to the Sri Lankan context.

Expected outcome:
A cleaned, structured, and normalized transaction dataset suitable for machine learning and reasoning tasks.

SO2: Develop a semantic transaction understanding model

To build and evaluate a machine learning model that can classify and semantically interpret personal financial transactions by identifying meaningful transaction categories and distinguishing among different types of financial inflows and transfers.

This objective goes beyond surface-level categorization by enabling the system to infer whether a transaction likely corresponds to salary, freelance income, gifts, reimbursements, loans, transfers between own accounts, capital-related receipts, or other financial events. The purpose is to establish the semantic understanding layer required before tax reasoning can take place.

Expected outcome:
A transaction semantic classification model with measurable predictive performance on annotated or weakly supervised transaction data.

Indicative evaluation measures:
Classification accuracy, macro F1-score, precision, recall, and robustness across different transaction formats.

SO3: Design an economic event identification and tax-rule mapping mechanism

To develop a reasoning layer that maps semantically interpreted transactions to underlying economic events and connects those events to relevant tax-rule categories for downstream taxability assessment.

This objective is central to the novelty of the proposed research. Instead of stopping at transaction classification, the framework should identify what the transaction economically represents and then determine which tax-relevant interpretation is most appropriate. For example, the system should be able to distinguish whether a recurring inflow is likely employment income, whether an isolated transfer may represent a gift or reimbursement, or whether a transaction should be treated as uncertain and referred for review.

Expected outcome:
A rule-grounded mapping mechanism that converts transaction semantics into tax-relevant economic interpretations.

SO4: Develop an explainable taxability reasoning and evidence generation framework

To design an explainable inference framework that determines whether a transaction is taxable, non-taxable, partially taxable, exempt, or uncertain, while generating a transparent reasoning trace and supporting evidence for each decision.

This objective ensures that the proposed system functions as a decision-support framework rather than a black-box predictor. The system should provide interpretable outputs that explain how a transaction was understood, what economic event was inferred, what tax-rule category was considered relevant, and why the final taxability conclusion was reached. This reasoning layer should be understandable to users, auditors, or researchers reviewing the decision.

Expected outcome:
An explainable taxability reasoning module capable of producing human-readable evidence chains for inferred decisions.

SO5: Estimate taxable income amounts with uncertainty awareness and calibrated confidence

To develop a taxable income inference mechanism that estimates the likely taxable amount associated with relevant transactions and produces calibrated confidence scores to indicate the reliability of each prediction.

This objective is important because not all financial inflows can be interpreted with equal certainty. The proposed framework should therefore not only produce a taxability outcome, but also quantify uncertainty so that ambiguous or low-confidence cases can be flagged for human review. This makes the system more realistic and trustworthy in practical tax-support settings.

Expected outcome:
A confidence-aware taxable amount estimation model capable of generating prediction outputs together with uncertainty indicators.

Indicative evaluation measures:
RMSE or MAE for taxable amount estimation, calibration error, confidence reliability, and uncertainty-based review flag quality.

SO6: Incorporate novelty detection and human-in-the-loop review support

To design a mechanism for identifying unfamiliar, ambiguous, or previously unseen transaction patterns and routing them for human review in order to improve model adaptability and long-term system reliability.

Financial transaction ecosystems are dynamic, and real-world data may contain patterns not represented in the original training set. This objective ensures that the framework can handle such cases responsibly by combining automated inference with expert correction when needed. The outputs of this review process may also support iterative refinement of the model over time.

Expected outcome:
A review-support mechanism for uncertain or novel transaction cases, enabling active learning and safer deployment.

SO7: Validate the proposed framework experimentally in the Sri Lankan tax-compliance context

To evaluate the effectiveness, reliability, and practical usefulness of the proposed framework using appropriate datasets, baselines, and performance metrics that reflect both predictive quality and decision-support value.

This objective ensures that the research is scientifically grounded and measurable. The evaluation should assess not only model accuracy, but also explainability, uncertainty behavior, generalization across different transaction formats, and the usefulness of the generated reasoning in reducing manual review effort.

Expected outcome:
A complete empirical validation of the framework supported by experiments, comparative analysis, and error analysis.

Indicative evaluation measures:
Macro/micro F1-score, accuracy, RMSE or MAE, expected calibration error, out-of-distribution generalization performance, and reduction in manual review burden during human evaluation.

2.3 Alignment of the Objectives with the Research Problem

The main objective and specific objectives are collectively designed to address the identified research problem: the absence of an explainable, context-aware, and intelligent framework for semantically analyzing personal financial transactions and inferring taxable income for personalized income tax compliance in Sri Lanka.

More specifically:

- **SO1** addresses the challenge of heterogeneous and noisy financial records.
- **SO2** addresses the lack of robust semantic understanding of transaction data.
- **SO3** addresses the missing link between transaction classification and tax-relevant economic interpretation.
- **SO4** addresses the lack of explainable taxability reasoning in current systems.
- **SO5** addresses uncertainty and the need for confidence-aware taxable amount inference.
- **SO6** addresses adaptability to unseen transaction types and the need for human oversight.
- **SO7** addresses the need for measurable scientific validation and practical decision-support evaluation.

Together, these objectives provide a complete and technically coherent roadmap for developing the proposed component as both a research contribution and a practical building block within the larger AI-driven tax compliance system.

3. METHODOLOGY

This section describes the methodological framework used to design, develop, and evaluate the proposed Financial Transaction Semantic Reasoning and Explainable Taxable Income Inference system. The methodology focuses on transforming raw financial transaction data into interpretable tax-related insights through a multi-stage processing pipeline that integrates machine learning, semantic interpretation, rule-based reasoning, and explainable inference.

The overall methodology follows a layered pipeline consisting of data acquisition, transaction preprocessing, semantic representation learning, economic event inference, taxability reasoning, and model evaluation. Each stage contributes to progressively transforming raw financial records into structured and explainable tax-relevant outputs.

3.1 Overall Methodological Framework

The proposed methodology is structured as a multi-stage financial transaction analysis and reasoning pipeline. Raw transaction records are first collected or generated from heterogeneous financial sources such as anonymized bank statements, synthetic financial records, and manually curated mock transaction data. These records are then normalized into a unified representation containing core fields such as transaction description, amount, date, transaction direction, and contextual metadata.

Next, the normalized transaction data are passed through a semantic understanding layer, where machine learning models analyze transaction text and context to infer the likely transaction meaning. The semantic outputs are then passed into an economic event identification stage, which attempts to determine whether the transaction corresponds to salary, freelance income, reimbursement, gift, loan, transfer between own accounts, capital-related receipt, or another event type. These economic event interpretations are then connected to a tax-rule mapping layer that determines the most relevant tax treatment category.

The final reasoning stage produces an explainable output containing the inferred taxability status, estimated taxable amount, calibrated confidence score, and supporting evidence. Transactions with low confidence or unfamiliar patterns are flagged for human review. The complete framework is then validated through classification, reasoning, calibration, and generalization experiments.

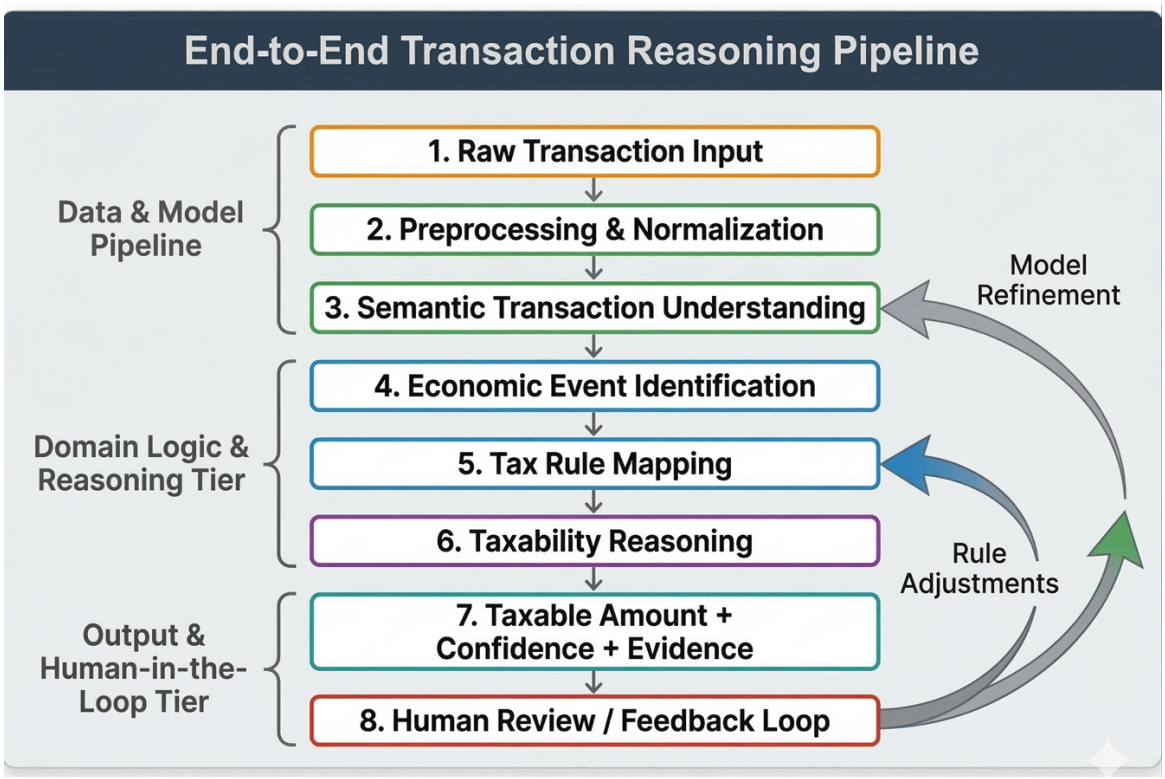


Figure 1 – End-to-End Transaction Reasoning Pipeline

3.2 Data Collection and Dataset Construction

One of the main challenges in this research is the scarcity of publicly available, labeled financial transaction datasets suitable for taxability reasoning. Therefore, the study will adopt a multi-source dataset construction strategy.

First, a **synthetic financial transaction generator** will be developed to create realistic transaction records reflecting different income and non-income scenarios. These synthetic records will vary by transaction description, amount, date, format, bank style, and transaction type. The generated transactions will represent cases such as salary deposits, freelance payments, reimbursements, gifts, loans, own-account transfers, and potentially ambiguous inflows.

Second, where feasible, a limited collection of **anonymized real or mock bank transaction samples** will be obtained through voluntary contributions, controlled examples, or manually created bank-style transaction statements. Personally identifiable information will be removed before use. Third, weak supervision techniques and expert-defined heuristics will be used to generate provisional labels where full manual annotation is not possible.

The final dataset will therefore consist of:

- synthetic transactions,
- weakly labeled transaction records,
- manually reviewed validation samples,
- and uncertainty / edge-case samples for human-in-the-loop evaluation.

3.3 Data Preprocessing and Normalization

Financial transaction data are typically noisy, inconsistent, and highly dependent on institution-specific formats. Therefore, before any semantic analysis can be performed, the raw records must undergo preprocessing and normalization.

This stage will include:

- extraction of relevant transaction fields,
- normalization of transaction descriptions,
- removal of irrelevant symbols and formatting noise,
- tokenization and encoding of transaction text,
- normalization of dates and amounts,
- creation of direction-of-flow indicators,
- and optional handling of multilingual or mixed-language descriptions.

Cross-bank normalization is particularly important because the same economic event may appear differently across different statement structures. The preprocessing stage will therefore aim to convert heterogeneous raw financial records into a consistent machine-readable schema that can be used reliably across models.

3.4 Semantic Transaction Understanding

The semantic transaction understanding stage focuses on identifying the meaning of financial transactions based on their textual descriptions and contextual attributes.

A transformer-based language model will be used to generate contextual embeddings of transaction descriptions. These embeddings capture semantic relationships between words and allow the model to recognize patterns associated with different financial events.

Possible model choices include:

- BERT
- FinBERT
- Domain-adapted transformer models

The semantic model will classify transactions into meaningful financial categories such as:

- Salary payments
- Freelance income
- Gifts
- Loans
- Reimbursements
- Transfers
- Capital gains

The classification model will be trained using labeled or weakly supervised datasets.

3.5 Economic Event Identification and Tax Rule Mapping

The semantic classification output alone is not sufficient for tax decision support. Therefore, the next stage of the methodology introduces an economic event identification layer that interprets what the transaction actually represents in a practical financial sense.

For example, a recurring monthly credit with consistent amounts may be interpreted as employment income, while an isolated family transfer may correspond to a gift or non-taxable personal transfer. Likewise, a refund-like description combined with previous debit context may suggest reimbursement rather than income. The identified economic event is then mapped to a set of tax-relevant categories or rule templates.

This tax-rule mapping mechanism does not attempt to replace legal interpretation fully. Instead, it functions as a structured decision-support layer that narrows possible tax treatments and prepares the inputs for explainable taxability reasoning.

3.6 Explainable Taxability Reasoning Framework

The core novelty of the proposed methodology lies in the taxability reasoning stage. In this stage, the framework generates a structured reasoning process that explains why a transaction is considered taxable, non-taxable, partially taxable, exempt, or uncertain.

The reasoning module will combine:

- semantic transaction interpretation,
- contextual sequence information,
- economic event inference,
- and tax-rule mapping outputs

to produce a human-readable decision trace. Rather than outputting only a final class label, the framework will produce an explanation chain that describes the evidence considered, the inferred economic event, the applicable reasoning basis, and the resulting taxability conclusion.

This module is intended to function as an explainable decision-support mechanism for end users, auditors, or tax professionals, making the system more transparent and practically useful than a black-box prediction model.

3.7 Uncertainty Estimation, Novelty Detection, and Human-in-the-Loop Review

Since financial transactions can be ambiguous, the methodology will include mechanisms for uncertainty estimation and novelty detection. This is important because some transactions may not fit confidently into any known category, especially when new abbreviations, unusual transaction patterns, or unseen financial behaviors appear.

The framework will therefore estimate calibrated confidence for each inferred decision and flag low-confidence or out-of-distribution cases for human review. Novel transaction discovery methods such as clustering or uncertainty-based anomaly detection may also be used to identify patterns not captured by the original training labels. Human feedback on flagged transactions can then be used to iteratively improve the system.

3.8 Model Development Strategy

The model development process will proceed in iterative stages. Initially, baseline models such as traditional machine learning classifiers or simple text embedding approaches will be implemented to establish a performance benchmark. These will then be compared against more advanced transformer-based and sequence-aware models.

The modeling workflow will include:

- baseline semantic classification models,
- advanced semantic models with contextual embeddings,
- optional sequence-aware models for transaction timelines,
- uncertainty estimation methods,
- and reasoning-layer integration.

Ablation studies will be conducted to examine the effect of different components, such as:

- with vs without sequence context,
- with vs without weak supervision,
- with vs without confidence calibration,
- with vs without tax-rule mapping.

This ensures that the final system is not only built, but scientifically evaluated in a way that demonstrates the contribution of each major design choice.

3.9 Tools and Platforms

The implementation will use tools and platforms selected according to research and system requirements rather than convenience alone. Python is the most suitable development language due to its strong ecosystem for natural language processing, machine learning, data preprocessing, and model evaluation. Libraries such as PyTorch, TensorFlow, scikit-learn, pandas, and Hugging Face Transformers are suitable for implementing the semantic classification and reasoning pipeline.

For API development and serving the trained models, a lightweight backend framework such as FastAPI may be used. For data storage and experiment tracking, structured datasets may be managed through local storage, relational databases, or document-based storage depending on the final system design. Visualization and interpretability tools may also be incorporated for confidence reporting and evidence display in the prototype.

The selection of these tools is justified by their compatibility with NLP workflows, support for experimental reproducibility, and ease of integration into a prototype decision-support system.

3.10 Model Evaluation and Validation

The proposed framework will be evaluated using both predictive and decision-support-oriented metrics. For semantic transaction classification, performance will be measured using classification accuracy, precision, recall, and macro F1-score. For taxable amount estimation, regression-based measures such as RMSE or MAE will be used. Since confidence-aware prediction is a major part of the framework, calibration quality will also be assessed using calibration-oriented measures such as expected calibration error.

Generalization performance will be tested across different transaction formats and bank styles to determine whether the framework remains effective under cross-bank variation. Ablation

experiments will be conducted to evaluate the contribution of context modeling, weak supervision, rule mapping, and uncertainty estimation. In addition, a small human-review based evaluation may be performed to assess whether the system reduces manual effort when compared with unaided review.

This validation strategy is important because the proposal guide clearly expects a methodology to specify **what metrics will be measured, what baseline will be used, and how the solution will be validated**.

Table 2 - Validation Metrics by Objective

Objective Area	Evaluation Type	Metrics
Semantic classification	Classification	Accuracy, Precision, Recall, Macro F1
Taxable amount inference	Regression	RMSE, MAE
Confidence quality	Calibration	ECE, confidence reliability
Cross-bank robustness	Generalization	OOD accuracy / F1
Human review support	Practical utility	Review effort reduction, uncertainty precision

3.11 Ethical Considerations

Because financial transaction data can contain sensitive personal information, this study must place strong emphasis on privacy, confidentiality, and responsible data handling. Any real or mock transaction data used in the study will be anonymized before analysis, and personally identifiable information such as account numbers, names, and reference identifiers will be removed or masked.

If voluntary or externally sourced transaction samples are used, informed consent and proper handling procedures will be applied where necessary. Synthetic data generation will be used extensively to reduce exposure to sensitive personal financial records. The system is intended as a decision-support tool rather than a legally binding automated tax authority, and therefore outputs should be interpreted as advisory or assistive rather than definitive legal determinations.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Component Architecture

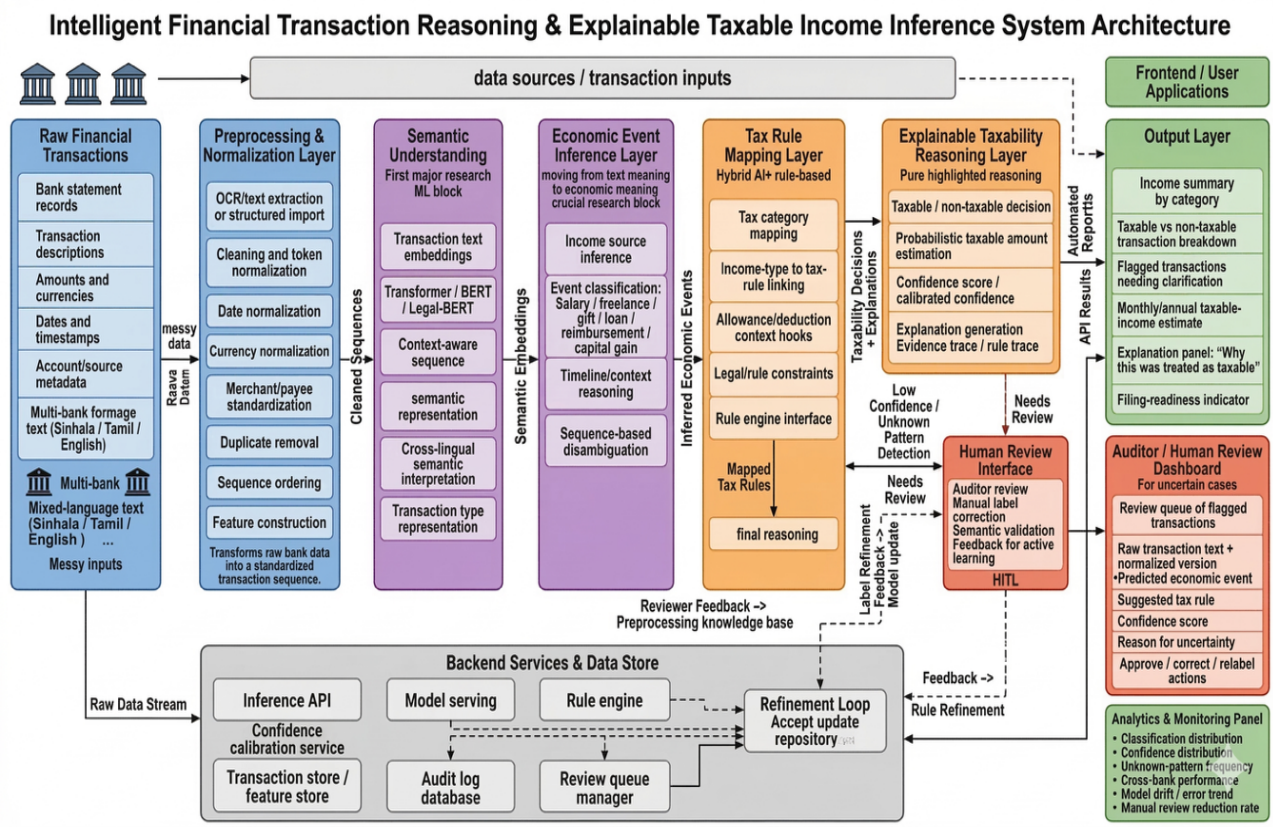


Figure 2 - System Architecture

This component focuses on **Financial Transaction Semantic Reasoning and Explainable Taxable Income Inference**. Its purpose is to transform raw personal financial transaction records into structured, tax-relevant outputs that can support personalized income tax compliance in Sri Lanka. At a high level, the component receives heterogeneous transaction records as input, processes and normalizes them, semantically interprets their meaning, identifies the likely underlying economic event, maps that event to tax-relevant categories, and produces an explainable taxability decision together with an estimated taxable amount, confidence score, and supporting evidence.

Unlike a basic transaction classification system, this component is designed as a **multi-stage reasoning architecture**. Each stage performs a distinct role, allowing the overall framework

to move from raw financial text and metadata toward an interpretable tax-support output. The architecture is modular so that individual layers can be improved, replaced, or extended without redesigning the full pipeline. This is important both for experimentation and for eventual integration with the other project components.

At the output level, the component can provide structured taxability information to downstream modules such as advisory, optimization, or personalized recommendation components in the wider project. However, within this proposal, the focus remains only on the internal architecture of this specific component.

4.2 Core Architectural Layers of the Component

The proposed component consists of several logically connected architectural layers, each responsible for a specific stage of transaction understanding and taxability inference.

4.2.1 Input Layer

The input layer receives raw financial transaction data from bank statements, mock financial records, or structured transaction feeds. These records may contain transaction descriptions, dates, amounts, debit or credit indicators, source-bank formatting, and other metadata. Since the incoming data may vary significantly in structure and quality, this layer primarily acts as the entry point for heterogeneous financial records.

4.2.2 Preprocessing and Normalization Layer

This layer converts raw transaction records into a unified internal schema. It performs text cleaning, extraction of relevant fields, normalization of numerical and temporal values, and cross-bank format standardization. The purpose of this layer is to ensure that downstream models operate on consistent, machine-readable data even when the original records differ in structure, wording, or notation.

4.2.3 Semantic Transaction Understanding Layer

The semantic understanding layer analyzes transaction descriptions and contextual attributes to infer the likely semantic category of the transaction. This stage focuses on understanding the practical meaning of the transaction text, rather than merely storing it. It may use transformer-based embeddings, contextual encoding, or other language-aware modeling approaches to identify semantic patterns in financial transactions.

4.2.4 Economic Event Identification Layer

This layer interprets the semantically processed transaction in terms of its likely real-world economic event. For example, it may infer whether a transaction corresponds to salary, freelance income, reimbursement, gift, loan receipt, internal transfer, or another event type. This stage is important because taxability depends not only on surface wording, but also on the underlying financial meaning of the event.

4.2.5 Tax Rule Mapping Layer

Once an economic event is inferred, the architecture maps that event to a relevant tax interpretation category. This layer acts as the bridge between machine-learned transaction understanding and tax-relevant reasoning. It does not attempt to replace formal legal interpretation, but rather provides structured support by narrowing the likely tax treatment of the transaction.

4.2.6 Explainable Taxability Reasoning Layer

This is the central reasoning layer of the component. It combines semantic interpretation, economic event inference, and tax-rule mapping to determine whether a transaction is likely taxable, non-taxable, partially taxable, exempt, or uncertain. It also generates a reasoning trace that explains how the conclusion was reached.

4.2.7 Output and Review Layer

The final layer produces the structured system output, which may include:

- inferred transaction meaning,
- identified economic event,
- suggested taxability status,
- estimated taxable amount,
- confidence level,
- supporting explanation or evidence.

If the confidence level is low or the transaction pattern appears unfamiliar, the case can be redirected to a human review path.

4.3 Internal Component Interactions

The internal architectural layers operate sequentially, but they are also logically interdependent. The preprocessing and normalization layer prepares structured transaction inputs for semantic interpretation. The semantic understanding layer then generates representations or labels that are consumed by the economic event identification layer. The inferred economic event is then passed to the tax rule mapping layer, which narrows the transaction into tax-relevant interpretations. These outputs are finally used by the explainable reasoning layer to generate a final taxability decision.

In addition to the forward flow, the architecture also includes a feedback-oriented interaction path. When the system encounters ambiguous, low-confidence, or previously unseen transaction patterns, the output layer may trigger a review request instead of producing a strong automated conclusion. This allows the architecture to balance intelligent automation with human oversight.

This interaction model is important because it supports both **modularity** and **traceability**. Each stage contributes a specific transformation of information, making the overall system easier to evaluate, refine, and explain.

4.4 Data Flow Through the Component

The data flow of the proposed component begins when raw transaction records enter the system. These records are first transformed into a normalized internal representation. The normalized records are then encoded and interpreted semantically, allowing the system to recognize transaction-level meaning. This semantic output is enriched with contextual information and passed into the economic event inference stage, where the likely real-world nature of the transaction is identified.

Once the event is inferred, the transaction is associated with one or more tax-relevant categories through the tax rule mapping stage. The reasoning engine then combines these outputs to generate a final taxability interpretation and taxable amount estimate. At this stage, the component also computes a confidence value and produces supporting evidence. The final output is then returned either as an automated decision-support result or as a flagged case for human review.

This architecture ensures that data are transformed progressively from **raw transaction form** into **interpretable tax-support output**, with each intermediate step contributing to the final reasoning chain.

4.5 Scalability Considerations

The proposed architecture is intentionally modular to support future scalability. Since each stage is logically separated, improvements can be introduced incrementally without requiring a full redesign. For example, the semantic understanding model can be replaced with a stronger language model, or the tax rule mapping layer can be extended with more detailed reasoning logic, while keeping the rest of the architecture unchanged.

The architecture is also scalable in terms of input variation. Because the normalization layer standardizes heterogeneous records into a common schema, the system can be extended to support additional bank formats or transaction sources over time. Similarly, the use of uncertainty-aware review routing allows the system to remain usable even when data complexity grows beyond the original training scope.

From a practical deployment perspective, the architecture can also support service-based separation, where preprocessing, inference, reasoning, and review functions are implemented as separate modules or APIs if needed at a later stage.

4.6 Security and Privacy Considerations

Because this component processes financial transaction data, privacy and security are essential architectural concerns. Sensitive personal information such as account identifiers, names, and bank-specific reference details should be anonymized or masked before the data enter the main inference pipeline. The architecture should minimize unnecessary retention of identifiable financial records and prioritize the use of anonymized or synthetic datasets during development and testing.

In a deployed setting, secure transmission of transaction records, controlled access to stored data, and proper audit logging should be considered part of the system design. The architecture should also ensure that explainability outputs do not expose unnecessary sensitive details while still remaining useful for decision support. These considerations are especially important because the component operates on high-sensitivity financial information.

5. USER REQUIREMENTS

5.1 Overview of User Requirements

The user requirements for this component are derived from the practical needs of users who interact with personal financial transaction data for income tax compliance purposes. The component is intended to support users by transforming raw financial transaction records into tax-relevant, explainable outputs. In this context, the primary users include individual taxpayers, tax advisors or auditors, and other internal project components that may consume structured transaction-level taxability outputs.

In line with the proposal guide, the requirements are defined so that the component addresses a real user need rather than functioning only as a technical experiment. The requirements therefore focus not only on correct transaction understanding, but also on usability-related qualities such as explainability, reliability, privacy, performance, and review support for uncertain cases.

Table 3 – Primary Users of the Component

Individual taxpayer	understands transaction taxability and taxable amount
Tax advisor / auditor	reviews evidence-backed reasoning and flagged cases
Downstream project component	consumes structured output for advisory or recommendation logic

5.2 Functional Requirements

The functional requirements describe what the component must do in order to fulfill its intended purpose.

Table 4 – Functional Requirements

FR1: Transaction Data Ingestion	The component shall accept raw financial transaction records from supported input sources such as structured transaction files, bank-statement-derived records, or normalized mock transaction datasets.

FR2: Transaction Field Extraction and Normalization	The component shall extract and normalize key transaction attributes including description, date, amount, direction of flow, and relevant metadata into a unified internal schema.
FR3: Semantic Transaction Classification	The component shall analyze transaction descriptions and contextual features to classify transactions into semantically meaningful categories.
FR4: Economic Event Identification	The component shall infer the likely underlying economic event represented by a transaction, such as salary, freelance income, gift, reimbursement, loan receipt, or internal transfer.
FR5: Tax Rule Mapping	The component shall map inferred economic events to relevant tax-reasoning categories to support taxability analysis.
FR6: Explainable Taxability Inference	The component shall determine whether a transaction is taxable, non-taxable, partially taxable, exempt, or uncertain, and generate an explainable reasoning trace for that decision.
FR7: Taxable Amount Estimation	The component shall estimate the taxable amount associated with relevant transactions where applicable.
FR8: Confidence Scoring	The component shall produce a confidence score or uncertainty indicator for each taxability-related output.
FR9: Novel or Uncertain Transaction Flagging	The component shall identify low-confidence, ambiguous, or previously unseen transaction patterns and flag them for human review.
FR10: Structured Output Generation	The component shall generate structured outputs containing transaction meaning, economic event, taxability status, taxable amount estimate, confidence, and supporting explanation.
FR11: Review Support Interface Output	The component shall provide flagged cases and reasoning evidence in a format suitable for manual review by a user or downstream system.

FR12: Inter-Component Output Readiness	The component shall expose its final structured outputs in a form that can be consumed by other project components, such as advisory or recommendation modules, where applicable.

5.3 Non-Functional Requirements

The non-functional requirements define the quality attributes, operational constraints, and system properties that the component must satisfy to be practically useful and technically reliable. The guide specifically suggests that you should be ready to explain your top non-functional requirements, so this subsection should be clear and concrete.

Table 5 – Non-Functional Requirements

NFR1: Performance	The component should process a transaction request and return a structured taxability output within an acceptable response time for prototype use.
NFR2: Accuracy and Reliability	The semantic classification and taxability inference outputs should achieve acceptable predictive performance according to the evaluation metrics defined in the methodology.
NFR3: Explainability	The component should provide outputs that are understandable to non-expert or semi-expert users by presenting a clear reasoning trace rather than only a raw prediction label.
NFR4: Privacy and Data Security	The component must protect sensitive financial transaction data through anonymization, controlled access, and secure handling during processing, storage, and evaluation.
NFR5: Scalability	The architecture should support extension to additional transaction formats, banks, and future rule categories without requiring major redesign.
NFR6: Maintainability	The component should be modular so that preprocessing, semantic modeling, tax-rule mapping, and reasoning layers can be updated independently.
NFR7: Usability of Review Outputs	The output format for flagged or uncertain transactions should be readable and practically useful for manual review.

NFR8: Robustness to Noisy Input	The component should remain functional when transaction descriptions contain abbreviations, formatting inconsistencies, or incomplete text.
NFR9: Interoperability	The component should produce outputs in a format that can be integrated with other modules of the broader project.
NFR10: Auditability	The component should preserve enough intermediate reasoning information to support later review, debugging, and result justification.

6. COMMERCIALIZATION PLAN

6.1 Commercialization Overview

The proposed component, Financial Transaction Semantic Reasoning and Explainable Taxable Income Inference, has strong commercialization potential as an AI-driven financial intelligence module for tax-compliance support. Its core commercial value lies in its ability to convert raw personal financial transactions into structured, explainable, and tax-relevant outputs. Unlike conventional bookkeeping tools or tax filing portals, the proposed component is intended to provide semantic transaction understanding, taxability reasoning, taxable amount estimation, and confidence-aware evidence generation.

From a commercialization perspective, this component can be positioned as a decision-support engine rather than a standalone tax filing platform. In practical terms, it can be offered as:

- a backend intelligence API for tax-tech platforms,
- an enterprise module for accounting or payroll software providers,
- a compliance support tool for tax consultants and audit teams,
- or a feature layer integrated into a larger personalized tax advisory system.

This approach is commercially attractive because the component solves a high-friction problem that currently depends heavily on manual interpretation of financial records.

6.2 Value Proposition

The main value proposition of the proposed component is that it reduces the effort, ambiguity, and inconsistency involved in interpreting financial transactions for tax-compliance purposes. It offers a more advanced capability than ordinary transaction categorization by generating an evidence-based reasoning path that explains **what a transaction means**, **which economic event it likely represents**, and **why it is considered taxable, non-taxable, partially taxable, or uncertain**.

The component creates value in several ways:

- It reduces the manual effort required to inspect large numbers of personal financial transactions.
- It improves consistency in transaction interpretation across different users or reviewers.
- It supports faster tax-compliance workflows by converting raw records into tax-relevant structured outputs.
- It provides confidence-aware outputs, allowing uncertain cases to be reviewed instead of forcing unreliable automation.
- It is especially relevant in local contexts such as Sri Lanka, where multilingual transaction descriptions and cross-bank variation make manual review difficult.

This value proposition is commercially meaningful because it supports both productivity improvement and decision quality enhancement.

6.3 Target Market

The commercialization target for this component should be defined at two levels: **primary market** and **secondary market**.

6.3.1 Primary Target Market

The primary target market consists of business and professional users who work with tax-relevant financial records and would benefit from automated transaction reasoning.

These include:

- **Tax consultants and tax advisory firms**
  Professionals who review individual income-related transactions and prepare tax-relevant interpretations for clients.
- **Accounting firms and bookkeeping service providers**
  Organizations that need faster transaction categorization and tax-relevant income identification support.
- **Payroll and HR software providers**
  Software vendors that may wish to integrate taxability reasoning features into salary and employee tax-support workflows.
- **Fintech and tax-tech platforms**
  Companies building digital financial management or tax-assistance solutions that need an explainable transaction intelligence engine.

6.3.2 Secondary Target Market

The secondary target market includes:

- Individual taxpayers with multiple income streams
- Small business owners managing mixed personal/business cash flows
- Audit and compliance support teams
- Institutions developing digital tax-support services

For the proposal stage, it is best to emphasize the B2B and professional market first, because that is easier to justify commercially than selling directly to the general public at the beginning.

6.4 Potential Customers and Users

Although the end users of the reasoning output may include taxpayers, the most likely paying customers in the early commercialization phase are organizations that handle financial interpretation at scale.

Likely paying customers

- accounting firms,
- tax consultancies,

- payroll software vendors,
- digital tax platforms,
- compliance-focused fintech providers.

Likely end users

- tax professionals,
- accountants,
- auditors,
- compliance officers,
- advanced individual taxpayers.

This distinction is important because the **user** and the **payer** may not always be the same.

6.5 Unique Selling Proposition (USP)

The proposed component's unique selling proposition is that it goes beyond standard financial transaction categorization and provides explainable taxability reasoning tailored for income tax compliance.

Its key differentiators are:

- Semantic understanding rather than simple keyword matching
- Economic event identification rather than flat category labeling
- Tax-rule-oriented reasoning support
- Confidence scoring and uncertainty-aware flagging
- Evidence generation for reviewer trust
- Suitability for noisy, heterogeneous, and potentially multilingual financial records
- Potential localization for Sri Lankan tax-compliance needs

In commercial terms, the system is not just another financial classifier. It is a transaction-to-tax reasoning engine.

6.6 Revenue Model

1. Subscription-based B2B licensing

Accounting firms, tax consultants, or software providers pay a monthly or annual subscription to use the reasoning engine.

2. API usage-based pricing

Organizations pay based on:

- number of processed transactions,
- number of users,
- or number of tax analyses performed.

3. Enterprise integration fees

Custom integration, local deployment, and workflow customization can generate one-time setup revenue.

4. Premium analytics / reporting features

Advanced dashboards, batch transaction review, and enhanced review workflows can be offered as premium capabilities.

7. BUDGET AND JUSTIFICATION

The proposed budget covers the resources required to design, develop, train, evaluate, and demonstrate the Financial Transaction Semantic Reasoning and Explainable Taxable Income Inference component within the planned research period. Since this component is primarily software- and data-driven, the budget focuses on computing resources, dataset preparation, experimentation support, and prototype demonstration rather than extensive hardware acquisition. Each cost item is directly linked to the successful implementation and validation of the component.

7.1 Estimated Budget

Table 6 – Estimated Budget for the Component

Resource Type	**Purpose**	**Estimated Amount(LKR)**
Cloud / GPU compute credits	Model training, fine-tuning, experimentation, and evaluation	45,000
Data storage and backup	Secure storage of datasets, intermediate model outputs, and experiment files	8,000

Internet and online research support	Downloading research papers, datasets, model packages, and cloud access	12,000
Software / API / productivity services	Supporting experimentation, collaboration, and prototype integration	10,000
Synthetic dataset generation and annotation support	Preparing structured transaction datasets, mock records, and labeling support	15,000
Testing and prototype demonstration expenses	Demo preparation, reporting, visualization, and evaluation material preparation	10,000
Contingency allocation	Unforeseen project expenses during implementation and testing	10,000
Total Estimated Budget		110,000

7.2 Budget Justification

Cloud / GPU compute credits

This allocation is required for training and evaluating machine learning models used in transaction semantic understanding, economic event identification, and taxable income inference. Since transformer-based or sequence-aware models can be computationally demanding, cloud-based compute resources provide a practical alternative to high-cost dedicated hardware.

Data storage and backup

This cost supports the secure storage of synthetic datasets, anonymized or mock transaction samples, trained models, experiment logs, and evaluation outputs. Backup storage is important to prevent data loss during iterative experimentation.

Internet and online research support

This allocation supports access to online datasets, research articles, software libraries, pretrained models, and cloud services used throughout the project. Since the component depends heavily on current ML tools and supporting literature, internet connectivity is a necessary operational expense.

Software / API / productivity services

Although much of the development can be carried out using open-source tools, certain supporting services may still be needed for workflow efficiency, experiment tracking, collaboration, or prototype integration. This amount also provides flexibility if limited paid tools or APIs become useful during implementation.

Synthetic dataset generation and annotation support

A major challenge of the proposed research is the scarcity of labeled financial transaction data. Therefore, resources are needed for synthetic transaction generation, mock record preparation, weak labeling support, and small-scale manual review or annotation work. This directly supports the dataset construction strategy defined in the methodology.

Testing and prototype demonstration expenses

This cost supports the preparation of evaluation artifacts, visualizations, demo material, reporting outputs, and presentation-ready prototypes needed to demonstrate the component's functionality and results.

Contingency allocation

A contingency amount is included to accommodate unexpected costs such as additional compute usage, extended storage needs, or testing-related expenses that may arise during the research process.

7.3 Budget Feasibility

The proposed budget is moderate and feasible for a software-centered research component. Unlike projects that require extensive sensor hardware, embedded devices, or specialized equipment, this component primarily depends on computing infrastructure and research support resources. The budget is therefore cost-efficient while still being sufficient to support model development, experimentation, evaluation, and demonstration.

7.4 Cost Optimization Considerations

To reduce unnecessary expenses, the project will prioritize:

- open-source software libraries and frameworks,
- free academic or research-tier tools where available,
- synthetic and weakly supervised data strategies instead of expensive large-scale manual labeling,
- and cloud-based computing only when local computing resources are insufficient.

This approach helps keep the budget realistic while maintaining the technical quality of the research.

8. WORK BREAKDOWN STRUCTURE (WBS)

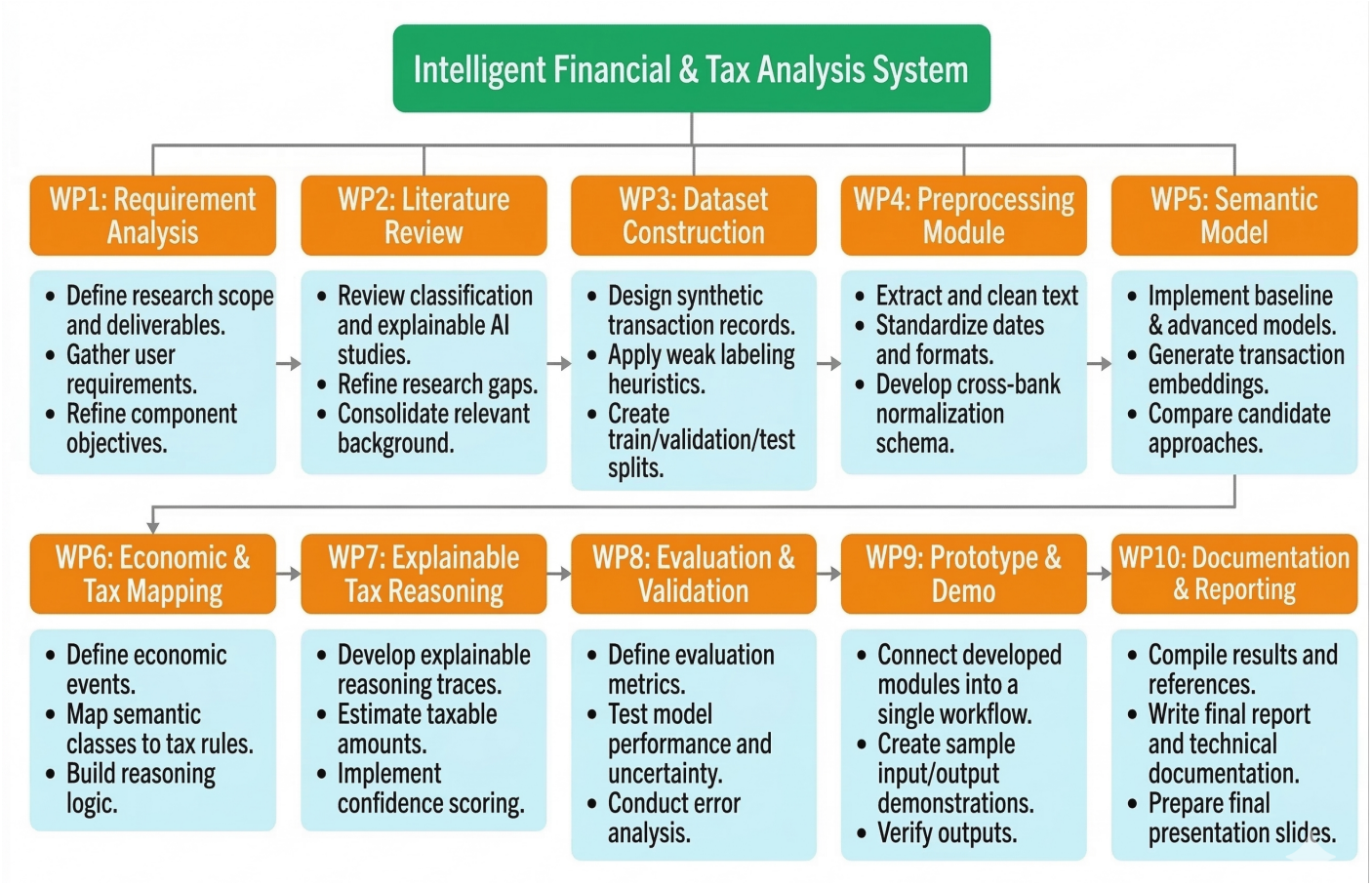


Figure 3 - Work breakdown structure and Deliverables

WP1: Requirement Analysis and Research Planning

This work package focuses on defining and refining the research scope, component boundaries, user requirements, objectives, and methodology. It ensures that the component is properly aligned with the proposal, research problem, and expected deliverables before implementation begins.

WP2: Literature Review and Gap Refinement

This work package covers the review of relevant literature related to financial transaction classification, explainable AI, uncertainty estimation, and related methods. It is intended to identify the technical and contextual gaps addressed by the component and to build a strong academic foundation for the study.

WP3: Dataset Construction and Preparation

This work package is responsible for creating and preparing the datasets required for training and evaluation. It includes synthetic transaction generation, mock or anonymized sample preparation, label design, and dataset splitting for model development.

WP4: Preprocessing and Normalization Module Development

This work package focuses on developing the preprocessing pipeline needed to convert heterogeneous transaction records into a consistent structured format. It supports downstream modeling by cleaning, standardizing, and organizing raw transaction data.

WP5: Semantic Transaction Understanding Model Development

This work package covers the development of the machine learning models used for semantic transaction analysis. It includes baseline and advanced model design, feature generation, training, tuning, and performance comparison.

WP6: Economic Event Identification and Tax Rule Mapping Development

This work package addresses the intermediate reasoning layer of the component by identifying underlying economic events and linking them to relevant tax categories. It helps bridge the gap between transaction classification and taxability analysis.

WP7: Explainable Taxability Reasoning and Confidence Module Development

This work package focuses on the core reasoning capability of the component. It includes generating explainable taxability decisions, estimating taxable amounts, producing evidence traces, and handling uncertainty through confidence scoring and review flagging.

WP8: Evaluation and Experimental Validation

This work package ensures the scientific validation of the proposed component through experiments and performance testing. It includes evaluating classification accuracy, taxable amount estimation, calibration quality, generalization ability, and model limitations.

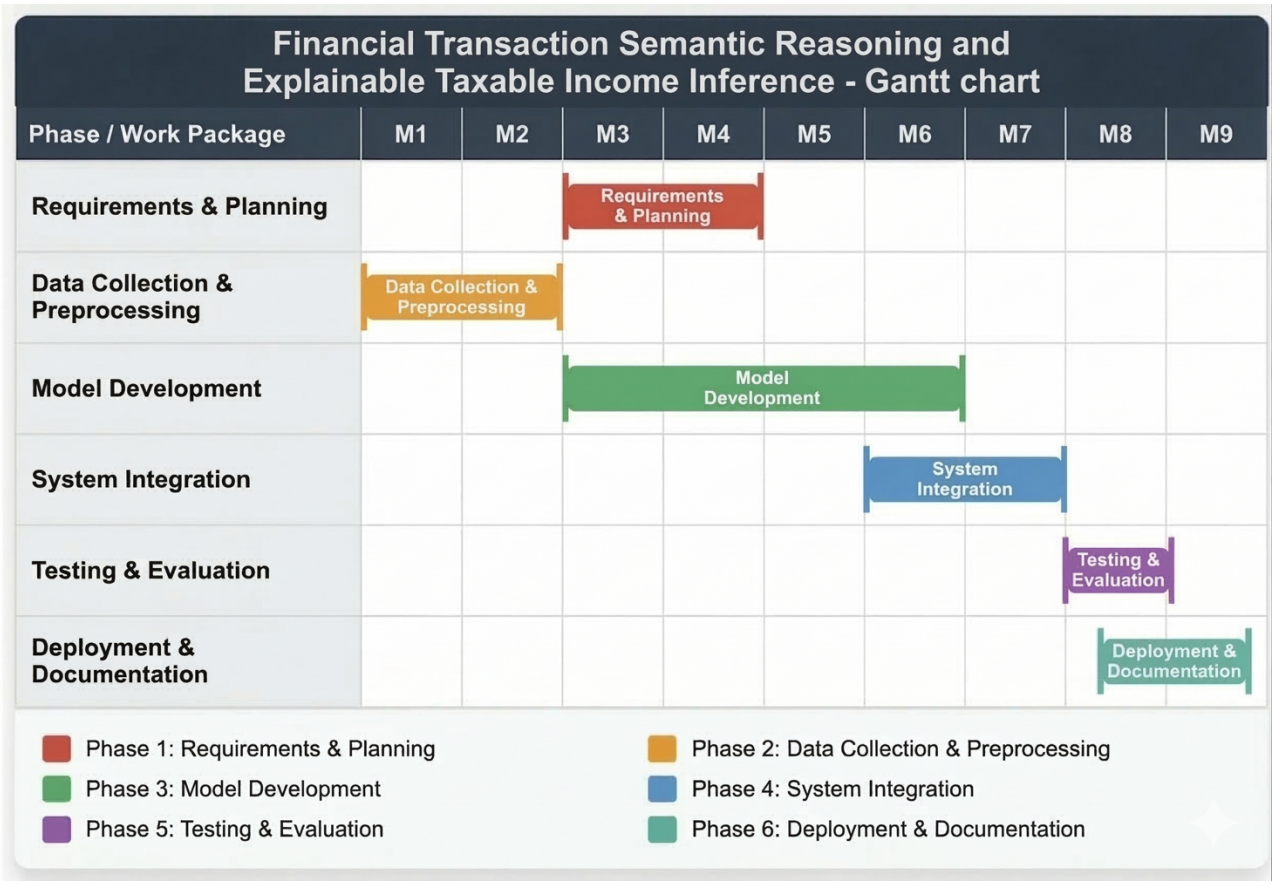
WP9: Prototype Integration and Demonstration

This work package is responsible for combining the developed modules into a usable prototype and preparing the component for demonstration. It includes sample input/output flows, interface preparation, and presentation-ready demo scenarios.

WP10: Documentation and Final Reporting

This final work package covers all report writing, technical documentation, and presentation preparation required to complete the research component. It ensures that the methodology, results, analysis, and deliverables are properly documented and communicated.

9. GANTT CHART



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LIST OF APPENDICES

Appendix A – High-Level Architecture of the Financial Transaction Semantic Reasoning Component

Appendix B – Work Breakdown Structure Diagram

Appendix C – Gantt Chart